MAT 21D Discussion 9 Notes

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March 05, 2026

§I Warm-Up

Let R be a region in the uv-plane. Discuss in groups, why a function of the form

$$\mathbf{r}(u,v) = f(u,v)\mathbf{i} + g(u,v)\mathbf{j} + h(u,v)\mathbf{k} \quad \text{where} \quad (u,v) \in R$$

represents a surface in $\mathbb{R}^3$.

Draw a picture illustrating the quantities

$$\mathbf{r}_u = \frac{\partial \mathbf{r}}{\partial u} \quad \text{and} \quad \mathbf{r}_v = \frac{\partial \mathbf{r}}{\partial v}$$

and show them to your group.

Solution. For each point (u,v) in R, the function $\mathbf{r}(u,v)$ gives us a point in $\mathbb{R}^3$ by evaluating the scalar functions $f(u,v)$, $g(u,v)$, and $h(u,v)$ and using them as the coefficients of the standard unit vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$. A surface worth of points in R will be mapped to a surface worth of points in $\mathbb{R}^3$, and in this way the function $\mathbf{r}(u,v)$ represents a surface in $\mathbb{R}^3$.

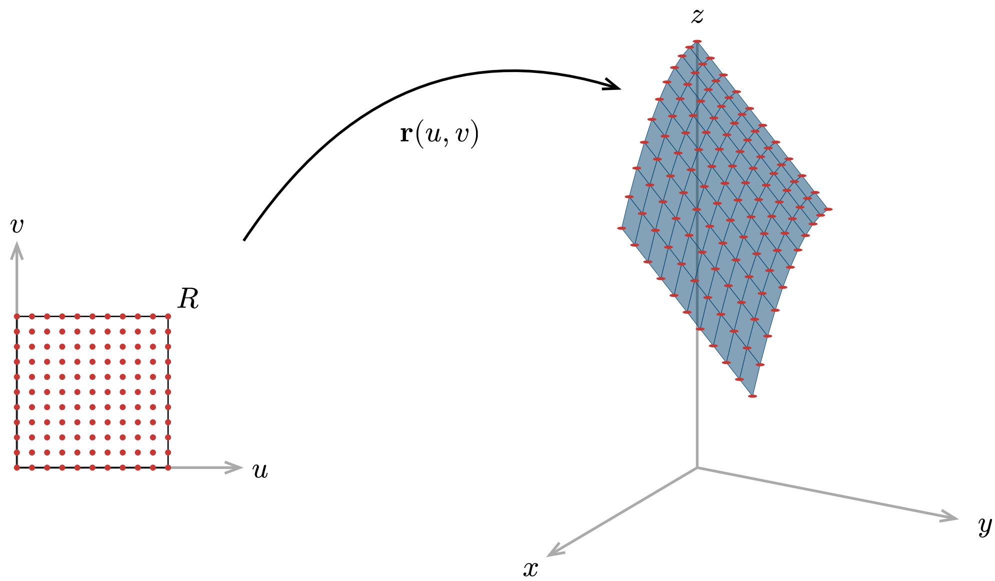

Figure 1: Illustration of the surface parameterized by $\mathbf{r}(u,v)$. Each point in R gets mapped to a point on the surface. A grid of points get mapped to a grid of points. A surface worth of points get mapped to a surface worth of points.

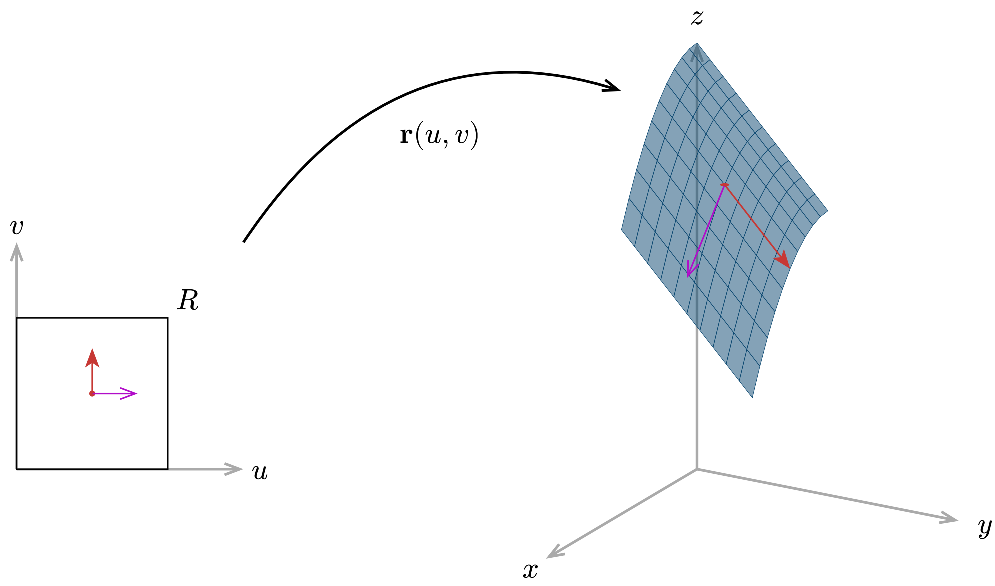

Figure 2: Illustration of the vectors $\mathbf{r}_u$ and $\mathbf{r}_v$ at a point on the surface. They are tangent to the surface and they point in the direction of how the image point $\mathbf{r}(u,v)$ will move if we increase u or v respectively. The tangent vector $\mathbf{r}_u$ is shown in purple, and the tangent vector $\mathbf{r}_v$ is shown in red.

§II Surface integral of a scalar function

If S is a surface parameterized by

$$\mathbf{r}(u,v) = f(u,v)\mathbf{i} + g(u,v)\mathbf{j} + h(u,v)\mathbf{k} \quad \text{where} \quad (u,v) \in R$$

for some region R in the uv-plane, then the surface integral of a scalar function $G(x,y,z)$ over S is given by

$$\iint_S G(x,y,z)\, d\sigma = \iint_R G(\mathbf{r}(u,v))|\mathbf{r}_u \times \mathbf{r}_v|\, du\, dv.$$

Here, $G(\mathbf{r}(u,v))$ is the expression obtained by substituting $x = f(u,v)$, $y = g(u,v)$, and $z = h(u,v)$ into $G(x,y,z)$.

Like the formula for line integrals over parameterized curves which implies that different parameterization of the same curve give the same line integral. The above formula for surface integrals implies that different parameterization of the same surface give the same surface integral. After all, the surface integral

$$\iint_S G(x,y,z)\, d\sigma$$

is only dependent on the surface S and the scalar function $G(x,y,z)$, and it makes sense regardless of whether S comes with a parameterization or not.

(Exercise 16.6.5) Let S be the surface parameterized by

$$\mathbf{r}(u,v) = u\mathbf{i} + v\mathbf{j} + (4-u-v)\mathbf{k} \quad \text{where} \quad (u,v) \in R$$

for the standard unit square R in the first quadrant of the uv-plane.

1. Draw a picture of the surface S.

2. Rewrite the surface integral

$$\iint_S z\, d\sigma$$

as an iterated integral in $du\, dv$.

3. Discuss with your group if the surface S is instead described as: the portion of the plane

$$x + y + z = 4$$

that lies above the square $0 \le x \le 1$, $0 \le y \le 1$ in the xy-plane, how would you derive the same parameterization as above?

Solution.

1. We can reexpress as the parameterization just by changing the variable names from u and v to x and y:

$$\mathbf{r}(x, y) = x\mathbf{i} + y\mathbf{j} + (4 - x - y)\mathbf{k} \quad \text{where} \quad 0 \le x \le 1, 0 \le y \le 1.$$

This makes it clear that the surface S is the graph of the function $h(x, y) = 4 - x - y$ over the square $0 \le x \le 1$, $0 \le y \le 1$ in the xy-plane. We can also figure out the corners of the surface S by plugging in the corners of the square R into the parameterization:

$$\mathbf{r}(0,0) = (0,0,4), \quad \mathbf{r}(1,0) = (1,0,3),$$
$$\mathbf{r}(0,1) = (0,1,3), \quad \mathbf{r}(1,1) = (1,1,2).$$

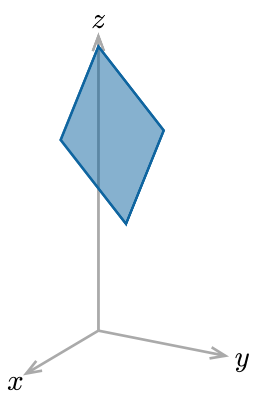

Figure 3: Picture of the surface S. The surface is a portion of the plane $x + y + z = 4$ that lies above the unit square R in the xy-plane. The corners of the surface are $(0, 0, 4)$, $(1, 0, 3)$, $(0, 1, 3)$, and $(1, 1, 2)$.

2. We need to express z in terms of u and v and compute $|\mathbf{r}_u \times \mathbf{r}_v|$. Looking at the parameterization, we see that

$$z = 4 - u - v$$

because $4 - u - v$ is $h(u, v)$ (the $\mathbf{k}$-component of $\mathbf{r}(u, v)$) in the above theorem. Next, we compute $\mathbf{r}_u$ and $\mathbf{r}_v$, which are

$$\mathbf{r}_u = \frac{\partial \mathbf{r}}{\partial u} = \mathbf{i} - \mathbf{k} \quad \text{and} \quad \mathbf{r}_v = \frac{\partial \mathbf{r}}{\partial v} = \mathbf{j} - \mathbf{k}.$$

Taking the cross product, we get

$$\mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{vmatrix} = \mathbf{i} + \mathbf{j} + \mathbf{k}.$$

I also recommend doing the long multiplication directly instead of using the determinant formula like so:

$$\begin{array}{cccc} & \mathbf{i} & & -\mathbf{k} \\ \times & & \mathbf{j} & -\mathbf{k} \\ \hline & & \mathbf{j} & \\ & \mathbf{i} & & +\mathbf{k} \\ \hline & \mathbf{i} & +\mathbf{j} & +\mathbf{k} \end{array}$$

Table 1: Long multiplication of $(\mathbf{i} - \mathbf{k}) \times (\mathbf{j} - \mathbf{k})$. The first row under the first line is the result of taking the cross product of $\mathbf{i}$ and $-\mathbf{k}$ with $-\mathbf{k}$, noting that $\mathbf{i} \times \mathbf{k} = -\mathbf{j}$ and $\mathbf{k} \times \mathbf{k} = 0$. The second row under the first line is the result of taking the cross product of $\mathbf{i}$ and $-\mathbf{k}$ with $\mathbf{j}$, noting that $\mathbf{i} \times \mathbf{j} = \mathbf{k}$ and $\mathbf{k} \times \mathbf{j} = -\mathbf{i}$. The final line is the sum of the two rows.

Thus, $|\mathbf{r}_u \times \mathbf{r}_v| = \sqrt{3}$. Therefore, the surface integral is

$$\iint_S z\, d\sigma = \iint_R (4 - u - v)\sqrt{3}\, du\, dv = \int_0^1 \int_0^1 (4 - u - v)\sqrt{3}\, du\, dv.$$

3. We will derive the same parameterization by solving for z in terms of x and y for points on the surface. The equation of the plane can be rearranged to get

$$z = 4 - x - y.$$

Thus, the surface S can be parameterized by

$$\mathbf{r}(x, y) = x\mathbf{i} + y\mathbf{j} + (4 - x - y)\mathbf{k} \quad \text{where} \quad 0 \le x \le 1, 0 \le y \le 1.$$

This is the same parameterization as above, just with different variable names.

Rewrite the surface integral

$$\iint_S (x - y - z)\, d\sigma$$

as iterated integrals for each of the following surfaces S.

1. The portion of the plane $x + y = 1$ in the first octant between $z = 0$ and $z = 1$.

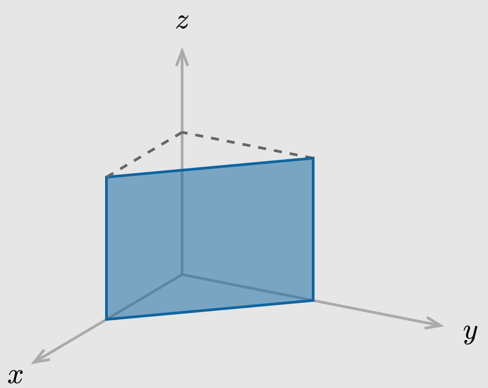

Figure 4: Picture of the portion of the plane $x + y = 1$.

> 2. The portion of the sphere $x^2 + y^2 + z^2 = 1^2$ in the first octant.

Solution.

1. While the surface S cannot be described as a graph of a function in x and y (does not pass the vertical line test), it can be described as a graph of a function in x and z (or in y and z). For example, the y value of a point on the surface is uniquely determined by x and z, and we can see this by rearranging the equation of the plane to get

$$y = 1 - x.$$

Thus, the surface S can be parameterized by

$$\mathbf{r}(x, z) = x\mathbf{i} + (1 - x)\mathbf{j} + z\mathbf{k} \quad \text{where} \quad 0 \leq x \leq 1, 0 \leq z \leq 1.$$

Next, we calculate

$$\mathbf{r}_x = \frac{\partial \mathbf{r}}{\partial x} = \mathbf{i} - \mathbf{j} \quad \text{and} \quad \mathbf{r}_z = \frac{\partial \mathbf{r}}{\partial z} = \mathbf{k}.$$

Therefore, we have

$$\mathbf{r}_x \times \mathbf{r}_z = (\mathbf{i} - \mathbf{j}) \times \mathbf{k} = \mathbf{i} \times \mathbf{k} - \mathbf{j} \times \mathbf{k} = -\mathbf{j} - \mathbf{i},$$

and so $|\mathbf{r}_x \times \mathbf{r}_z| = \sqrt{2}$. Finally, the surface integral is

$$\iint_S (x - y - z)\, d\sigma = \int_0^1 \int_0^1 (x - (1 - x) - z)\sqrt{2}\, dx\, dz.$$

2. One way is to parameterize the sphere is with spherical coordinates

$$\mathbf{r}(\varphi, \theta) = \sin(\varphi)\cos(\theta)\mathbf{i} + \sin(\varphi)\sin(\theta)\mathbf{j} + \cos(\varphi)\mathbf{k} \quad \text{where} \quad 0 \leq \varphi \leq \frac{\pi}{2}, 0 \leq \theta \leq \frac{\pi}{2}.$$

Next, we calculate

$$\mathbf{r}_\varphi = \frac{\partial \mathbf{r}}{\partial \varphi} = \cos(\varphi)\cos(\theta)\mathbf{i} + \cos(\varphi)\sin(\theta)\mathbf{j} - \sin(\varphi)\mathbf{k},$$
$$\mathbf{r}_\theta = \frac{\partial \mathbf{r}}{\partial \theta} = -\sin(\varphi)\sin(\theta)\mathbf{i} + \sin(\varphi)\cos(\theta)\mathbf{j}.$$

Taking the cross product, we get

$$\begin{array}{rccc} & \cos(\varphi)\cos(\theta)\mathbf{i} & +\cos(\varphi)\sin(\theta)\mathbf{j} & -\sin(\varphi)\mathbf{k} \\ \times & -\sin(\varphi)\sin(\theta)\mathbf{i} & +\sin(\varphi)\cos(\theta)\mathbf{j} & \\ \hline & \sin^2(\varphi)\cos(\theta)\mathbf{i} & & +\cos(\varphi)\sin(\varphi)\cos^2(\theta)\mathbf{k} \\ & & \sin^2(\varphi)\sin(\theta)\mathbf{j} & +\cos(\varphi)\sin(\varphi)\sin^2(\theta)\mathbf{k} \\ \hline & \sin^2(\varphi)\cos(\theta)\mathbf{i} & +\sin^2(\varphi)\sin(\theta)\mathbf{j} & +\cos(\varphi)\sin(\varphi)\mathbf{k} \end{array}$$

and so $\mathbf{r}_\varphi \times \mathbf{r}_\theta = \sin^2(\varphi)\cos(\theta)\mathbf{i} + \sin^2(\varphi)\sin(\theta)\mathbf{j} + \cos(\varphi)\sin(\varphi)\mathbf{k}$. Thus, we have

$$\begin{aligned} |\mathbf{r}_\varphi \times \mathbf{r}_\theta| &= \sqrt{\sin^4(\varphi)\cos^2(\theta) + \sin^4(\varphi)\sin^2(\theta) + \sin^2(\varphi)\cos^2(\varphi)} \\ &= \sqrt{\sin^4(\varphi) + \sin^2(\varphi)\cos^2(\varphi)} \\ &= \sin(\varphi). \end{aligned}$$

Finally, the surface integral is

$$\iint_S (x - y - z)\, d\sigma = \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} (\sin(\varphi)\cos(\theta) - \sin(\varphi)\sin(\theta) - \cos(\varphi))\sin(\varphi)\, d\varphi\, d\theta.$$

§III Surface integral of a vector field

If S is a (oriented) surface parameterized by

$$\mathbf{r}(u,v) = f(u,v)\mathbf{i} + g(u,v)\mathbf{j} + h(u,v)\mathbf{k} \quad \text{where} \quad (u,v) \in R$$

for some region R in the uv-plane, then the surface integral of a vector field $\mathbf{G}(x,y,z)$ is given by

$$\iint_S \mathbf{G}(x,y,z) \cdot \mathbf{n}\, d\sigma = \iint_R \mathbf{G}(\mathbf{r}(u,v)) \cdot (\pm(\mathbf{r}_u \times \mathbf{r}_v))\, du\, dv.$$

The plus or minus sign is chosen so that $\mathbf{n}$ points in the same direction as $\pm(\mathbf{r}_u \times \mathbf{r}_v)$. Also, $\mathbf{G}(\mathbf{r}(u,v))$ is the expression obtained by substituting $x = f(u,v)$, $y = g(u,v)$, and $z = h(u,v)$ into $\mathbf{G}(x,y,z)$.

The surface integral of a vector field gets us a step closer to completing the analysis of integrations on 3-dimensional space.

$$\begin{tikzcd}
\text{Scalar functions } f \arrow[r, "\nabla"] & \text{Vector fields } \mathbf{F} \arrow[r, "\nabla\times"] & \text{Vector fields } \mathbf{G} \arrow[r, "\nabla\cdot"] & \text{Scalar functions } g
\end{tikzcd}$$

$$\begin{tikzcd}
f(B) - f(A) \arrow[r, leftrightarrow, "\text{FT}"] & \displaystyle\int_C \mathbf{F} \cdot d\mathbf{r} \arrow[r, leftrightarrow, "\boxed{??}"] & \displaystyle\iint_S \mathbf{G} \cdot \mathbf{n}\, d\sigma \arrow[r, leftrightarrow, "\boxed{??}"] & \displaystyle\iiint_D g(x,y,z)\, dV
\end{tikzcd}$$

$$\begin{tikzcd}
\text{Points } A, B & \text{Curves } C \arrow[l, "\text{End points}"'] & \text{Surfaces } S \arrow[l, "\text{Boundary}"'] & \text{Solid regions } D \arrow[l, "\text{Boundary}"']
\end{tikzcd}$$

Let S be the portion of the plane $x + y = 1$ in the first octant between $z = 0$ and $z = 1$ in the direction away from the z-axis (the same surface as in the previous exercise). Let

$$\mathbf{G}(x,y,z) = x\mathbf{i} + y\mathbf{j}.$$

Let $\mathbf{r}(u,v)$ be the parameterization of S that you found in the previous exercise (you may use different symbols than u and v).

1. Use picture and **not calculation** to decide which sign to use so that $\pm(\mathbf{r}_u \times \mathbf{r}_v)$ points in the same direction as $\mathbf{n}$. Discuss with your group if you do not know how to do it with pictures.
2. Find $\mathbf{G}(\mathbf{r}(u,v))$.
3. Calculate the surface integral

$$\iint_S \mathbf{G}(x,y,z) \cdot \mathbf{n}\, d\sigma.$$

Solution. We can just reuse the parameterization

$$\mathbf{r}(x,z) = x\mathbf{i} + (1-x)\mathbf{j} + z\mathbf{k} \quad \text{where} \quad 0 \le x \le 1, 0 \le z \le 1.$$

1. The following figure shows $\mathbf{r}_x$ and $\mathbf{r}_z$ and their cross product $\mathbf{r}_x \times \mathbf{r}_z$.

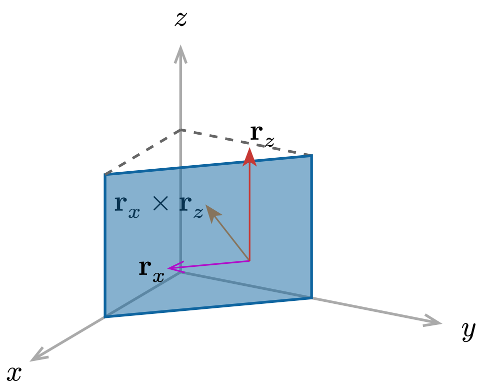

Figure 6: The tangent vector $\mathbf{r}_x$ is found by letting x increase while keeping z fixed, and the tangent vector $\mathbf{r}_z$ is found by letting z increase while keeping x fixed. Their cross product $\mathbf{r}_x \times \mathbf{r}_z$ is found by the right hand rule.

We see that $\mathbf{r}_x \times \mathbf{r}_z$ points in the direction towards the z-axis, and so it points in the opposite direction of $\mathbf{n}$, which points away from the z-axis. Therefore, the sign should be $-(\mathbf{r}_x \times \mathbf{r}_z)$.

2. We have

$$\mathbf{G}(\mathbf{r}(x, z)) = x\mathbf{i} + (1 - x)\mathbf{j}.$$

3. We have already computed $\mathbf{r}_x \times \mathbf{r}_z$ in the previous exercise, and it is $-\mathbf{j} - \mathbf{i}$. Therefore, we have

$$\mathbf{G}(\mathbf{r}(x, z)) \cdot (-(\mathbf{r}_x \times \mathbf{r}_z)) = (x\mathbf{i} + (1 - x)\mathbf{j}) \cdot (\mathbf{j} + \mathbf{i}) = x + (1 - x) = 1.$$

Finally, the surface integral is

$$\iint_S \mathbf{G}(x, y, z) \cdot \mathbf{n}\, d\sigma = \int_0^1 \int_0^1 1\, dx\, dz = 1.$$